EXPERIMENT NO: 7

DEVELOP A SIMPLE CALCULATOR USING XML-RPC

AIM

To develop a simple calculator where arithmetic operations are performed on a server and called by a client using XML-RPC.

ALGORITHM

1. Create a server program and register functions for add, subtract, multiply, divide.
2. Start the server on a fixed port.
3. Create a client program that connects to the server using ServerProxy.
4. Get input from the user and call the required function on the server.
5. Display the result returned by the server.

PROGRAM - SERVER (calc_server.py)

```
"""Calculator RPC server - exposes arithmetic functions over XML-RPC."""
import math
from xmlrpc.server import SimpleXMLRPCServer, SimpleXMLRPCRequestHandler

HOST, PORT = "localhost", 8000


class QuietHandler(SimpleXMLRPCRequestHandler):
    rpc_paths = ("/RPC2",)


def add(a, b):
    return a + b


def subtract(a, b):
    return a - b


def multiply(a, b):
    return a * b


def divide(a, b):
    if b == 0:
        raise ZeroDivisionError("Division by zero is not allowed")
    return a / b


def power(a, b):
    return a ** b


def square_root(a):
    if a < 0:
        raise ValueError("Square root of a negative number")
    return math.sqrt(a)
```

```
with SimpleXMLRPCServer((HOST, PORT), requestHandler=QuietHandler,
allow_none=True) as server:
    server.register_introspection_functions()
    for fn in (add, subtract, multiply, divide, power, square_root):
        server.register_function(fn)
    print(f"Calculator server running on http://{HOST}:{PORT}/RPC2")
    print("Press Ctrl+C to stop.")
    try:
        server.serve_forever()
    except KeyboardInterrupt:
        print("\nServer stopped.")
```

PROGRAM - CLIENT (calc_client.py)

```
"""Calculator RPC client - menu driven, calls the remote functions."""
import xmlrpc.client

proxy = xmlrpc.client.ServerProxy("http://localhost:8000/RPC2")

MENU = """
===== XML-RPC CALCULATOR =====
1. Add          2. Subtract
3. Multiply     4. Divide
5. Power        6. Square root
0. Exit
"""

binary = {"1": ("add", "+"), "2": ("subtract", "-"), "3": ("multiply", "*"),
          "4": ("divide", "/"), "5": ("power", "^")}

try:
    print("Methods available on server:", ", ".join(m for m in
proxy.system.listMethods()
                                                    if not m.startswith("system.")))
except ConnectionRefusedError:
    print("Cannot reach the server. Start calc_server.py first.")
    raise SystemExit(1)

while True:
    print(MENU)
    choice = input("Enter choice: ").strip()
    if choice == "0":
        print("Goodbye!")
        break
    try:
        if choice in binary:
            name, sym = binary[choice]
            x = float(input("Enter first number : "))
            y = float(input("Enter second number: "))
            result = getattr(proxy, name)(x, y)
            print(f"Result: {x} {sym} {y} = {result}")
        elif choice == "6":
            x = float(input("Enter number: "))
            print(f"Result: sqrt({x}) = {proxy.square_root(x)}")
        else:
            print("Invalid choice, try again.")
    except ValueError:
```

```
        print("Please enter numbers only.")
    except xmlrpc.client.Fault as err:
        print(f"Server error: {err.faultString}")
```

OUTPUT

```
Methods available on server: add, divide, multiply, power, square_root, subtract

===== XML-RPC CALCULATOR =====
1. Add          2. Subtract
3. Multiply     4. Divide
5. Power        6. Square root
0. Exit

Enter choice: 1
Enter first number : 12
Enter second number: 5
Result: 12.0 + 5.0 = 17.0

===== XML-RPC CALCULATOR =====
1. Add          2. Subtract
3. Multiply     4. Divide
5. Power        6. Square root
0. Exit

Enter choice: 4
Enter first number : 22
Enter second number: 7
Result: 22.0 / 7.0 = 3.142857142857143

===== XML-RPC CALCULATOR =====
1. Add          2. Subtract
3. Multiply     4. Divide
5. Power        6. Square root
0. Exit

Enter choice: 4
Enter first number : 9
Enter second number: 0
Server error: <class 'ZeroDivisionError'>:Division by zero is not allowed

===== XML-RPC CALCULATOR =====
1. Add          2. Subtract
3. Multiply     4. Divide
5. Power        6. Square root
0. Exit

Enter choice: 5
Enter first number : 2
Enter second number: 10
Result: 2.0 ^ 10.0 = 1024.0

===== XML-RPC CALCULATOR =====
1. Add          2. Subtract
3. Multiply     4. Divide
5. Power        6. Square root
0. Exit
```

```
Enter choice: 6
Enter number: 144
Result: sqrt(144.0) = 12.0

===== XML-RPC CALCULATOR =====
1. Add          2. Subtract
3. Multiply     4. Divide
5. Power        6. Square root
0. Exit

Enter choice: 6
Enter number: -9
Server error: <class 'ValueError'>:Square root of a negative number

===== XML-RPC CALCULATOR =====
1. Add          2. Subtract
3. Multiply     4. Divide
5. Power        6. Square root
0. Exit

Enter choice: 0
Goodbye!
```

RESULT

A simple calculator was developed using XML-RPC and tested successfully.